Specular Bit Architecture (SBA) for Bio-Photonic Neuromorphic Chips and DNA Storage: A Mathematical and Scientific Framework

Marco Oppido, Independent researcher scaccomarco@gmail.com

Abstract—This manuscript presents a substrate-agnostic, mathematically rigorous framework for the Specular Bit Architecture (SBA), conceived as a foundational informational layer for future bio-photonic neuromorphic processors and DNA data storage systems. The document focuses exclusively on the formal scientific properties of this dual-stream specular encoding—where information is generated simultaneously in two opposite directions as a unified vectorial state. We detail the local exact rewrite rules, deterministic sanitization and conflict-resolution policies, a lexicographic termination measure, and an informational mass–energy bookkeeping model consistent with an $E = Mc^2$ equivalence at the register level. The manuscript is explicit about the distinction between the informational model and physical claims: the quantities m_0 and c appearing in the bookkeeping are model parameters used for dimensional accounting; any empirical identification with physical constants requires independent experimental calibration. Without prescribing specific physical hardware, this work provides the exact definitions, invariants, and operational constraints that any underlying implementation must satisfy to guarantee logical correctness, strict locality, and bounded error propagation.

Index Terms—bio-photonic neuromorphic chips, bounded propagation, DNA data storage, Fibonacci positional weights, informational mass–energy, sanitization policy, signed-digit encoding, Specular Bit Architecture, Zeckendorf representation.

1 Introduction

Emerging high-density and low-latency substrates (photonic neuromorphic arrays, hybrid bio-electronic microcircuits, DNA archival media) motivate an informational layer that maximizes locality, tolerates transient ambiguous states, and provides deterministic, reproducible semantics for local correction. The Specular Bit Architecture (SBA) introduces a paradigm where information is not a simple scalar, but is generated simultaneously in two specular directions, forming a unified vectorial state.

SBA encodes signed information using *specular carriers* (mirror configurations) rather than an explicit sign bit, employs Fibonacci-derived positional weights to enable exact local rewrites, and defines deterministic sanitization and reconstruction policies that guarantee local termination and bounded propagation. This manuscript formalizes SBA as a mathematical framework and operational specification intended as a reference model for designers and researchers. The presentation separates formal informational claims (theorems, lemmas, operational protocols) from speculative physical interpretations; the latter are explicitly labeled and do not affect the formal correctness of the informational framework. At the communication-theoretic level, the framework is compatible with classical notions of reliable symbolic encoding and channel robustness [15]–[17].

2 Related Work

SBA builds on classical signed-digit arithmetic and Fibonacci positional systems [1]–[5], [18]. For substrate context we reference DNA data storage and coding literature [6], [7], [19]–[23], biological proofreading mechanisms [8], [13], and neuromorphic hardware reviews and hardware exemplars [9], [24]–[27]. The informational mass–energy bookkeeping is presented as a register-level modeling device and is not a claim about fundamental physics; it is inspired by thermodynamic interpretations of information processing and computation [11], [12], [28]–[30]. The dissipative interpretation adopted in the appendices is aligned with non-equilibrium self-organization frameworks [10], while the substrate-agnostic computational framing is consistent with foundational simulation perspectives [14].

3 Notation and Physical Carrier Layout

We present the SBA informational primitives and the carrier semantics that implementations must respect.

Definition 3.1 (Physical carrier per position)**.** *For each sequence position i we define two logical SBA channels A and B. Each channel is represented by a two-state physical carrier (two distinguishable physical states or two physical bits):*

$$\textit{Channel A: } (p_i^{(A)}, n_i^{(A)}), \qquad \textit{Channel B: } (p_i^{(B)}, n_i^{(B)}).$$

The logical digit decoded from a carrier relies on the differential balance between a positive pole (p) and a negative pole (n):

$$\delta(p, n) := p - n \in \{-1, 0, +1\},$$

with valid carrier patterns $00, 01, 10$. The pattern 11 is reserved as a transient ambiguous microstate and must be sanitized deterministically.

Remark 3.2 (Substrate mapping for DNA)**.** *In the context of DNA data storage, the four physical bits required for each positional index i map naturally to the four nucleobases. A*

biologically inspired mapping interprets base pairs as specular carriers: for instance, the presence or absence of specific bases (e.g., Adenine/Thymine for Channel A, Cytosine/Guanine for Channel B) can physically encode the (p, n) vectors. This explicitly models DNA information not as a scalar string, but as a dual-directional, specularly generated vectorial state.

3.1 Notation and Glossary

For reader convenience we collect the principal symbols used throughout the manuscript:

- $W_i^{(X)}$: positional weight at index i for channel $X \in \{A, B\}$.
- $a_i^{(X)}, b_i^{(X)}$: canonical digits in channel X.
- $d_i^{(X)}$: intermediate digit sum $a_i^{(X)} + b_i^{(X)}$.
- S: sanitization operator mapping transient $11 \mapsto 00$.
- σ: specular swap operator on carriers.
- $R_i^{(X)}$: deterministic reconstruction function at index i.
- $K_i^{(X)}$: sanitization flag at index i.
- E_{rad}: radiative energy register.
- m_0: model unit mass per unit positional weight.
- $R(d), m(d), p(d)$: lexicographic measure components (rightmost overflow index, multiplicity, pair).
- o_A, o_B: integer offsets for channel weights.

4 Fibonacci Positional Weights and Algebraic Identity

Let $(F_k)_{k\geq 0}$ denote the Fibonacci sequence with $F_0 = 0, F_1 = 1$. For each channel X fix an integer offset $o_X \geq 0$ and define positional weights

$$W_i^{(X)} := F_{i+o_X}, \qquad i \geq 0. \tag{1}$$

Notation convention: when the channel is clear from context, we write W_i instead of $W_i^{(X)}$.

For $i \geq 2$ these weights satisfy the identity

$$2W_i = W_{i+1} + W_{i-2}, \tag{2}$$

which is the algebraic basis for the exact local rewrite rules used to normalize intermediate digit sums (cf. (1) and (2)).

Remark 4.1 (Biological interpretation of weights)**.** *When applied to macromolecular substrates like DNA, these positional Fibonacci weights represent the hierarchical structural significance of the sequence. Instead of assigning arbitrary numerical values to bases, the geometric properties of the molecule rigidly define the positional weights, confining biological proofreading (mismatch repair) to a mathematically exact local neighborhood based on the identity $2W_i = W_{i+1} + W_{i-2}$ [8], [13].*

5 Canonical Form and Uniqueness

Definition 5.1 (Canonical sequence)**.** *A digit sequence $x = (x_i)$ with $x_i \in \{-1, 0, 1\}$ is* canonical *if for every i it holds $x_i \cdot x_{i+1} = 0$ (no adjacent nonzero digits).*

Lemma 5.2 (Zeckendorf uniqueness for positive support)**.** *For weights $W_i = F_{i+o}$ and digits $x_i \in \{0, 1\}$ with the non-adjacency constraint, the representation of any nonnegative integer as $\sum_i x_i W_i$ is unique (Zeckendorf theorem).*

Lemma 5.3 (Signed canonical uniqueness via specular carriers)**.** *Let c be a canonical carrier sequence. Define the specular swap operator σ that exchanges p_i and n_i for each carrier. Then $\delta(\sigma(c_i)) = -\delta(c_i)$ and σ is involutive. If two canonical carrier sequences have the same weighted value $\sum_i \delta(c_i)W_i$, then they coincide pointwise; hence the signed canonical representation is unique.*

Remark 5.4 (Informational CPT correspondence)**.** *The specular swap σ is formally analogous to an informational charge conjugation. If $\mathcal{P}$ denotes index reversal and $\mathcal{T}$ denotes reversal of synchronous round order, then the combined transformation $\sigma \circ \mathcal{P} \circ \mathcal{T}$ maps canonical informational histories to canonical histories preserving the lexicographic termination properties. This is a representational symmetry of the informational dynamics; it is not a claim about physical CPT invariance of spacetime.*

Proposition 5.5 (Uniqueness of dual independent streams)**.** *Let $W_i^{(A)} = F_{i+o_A}$ and $W_i^{(B)} = F_{i+o_B}$. Under the canonical non-adjacency constraint applied independently to each channel, the weighted representation is unique per channel. Consequently, the two-component logical state $\mathbf{V} = [V_A\ V_B]^T$ is unique.*

6 Intermediate Representation and Exact Local Rewrites

Given two canonical operands $A^{(X)}$ and $B^{(X)}$ for channel X, define the intermediate digit array in (3)

$$d_i^{(X)} := a_i^{(X)} + b_i^{(X)} \in \{-2, -1, 0, 1, 2\}. \tag{3}$$

6.1 Execution Model: Round Semantics

Normalization proceeds independently per channel via synchronous rounds with the following ordered phases:

1) **Read phase:** read current logical digits $d^{(X)}$ (pre-round snapshot).
2) **Compute phase:** compute interior rewrite updates for indices with $|d_i^{(X)}| = 2$ (local computations based on the Fibonacci identity).
3) **Write phase:** commit the computed updates to $d^{(X)}$ simultaneously (read-then-write semantics).
4) **Sanitization phase:** apply operator S to physical carriers; set flags $K_i^{(X)}$ for $11 \mapsto 00$ events.
5) **Scheduling & Arbitration:** for each flagged index, schedule a compensation candidate; apply Protocol C deterministic arbitration to select at most one committed injection per target.
6) **Commit phase:** commit the selected compensating injections and proceed to the next round.

Virtual guard cells $d_{-2} = d_{-1} = d_n = d_{n+1} = 0$ are assumed to enforce boundary semantics.

6.2 Formal Memory and Concurrency Model

Definition 6.1 (Round consistency model)**.** *A synchronous round satisfies* round-sequential consistency *if:*

1) ***Snapshot read:*** *the Read phase observes a single pre-round state s_t for all indices.*
2) ***Write linearization point:*** *all updates computed in the Compute phase are buffered and become visible atomically at one global linearization point τ_W at the end of the Write phase; no intermediate write is visible before τ_W.*
3) ***Atomic target lock before commit:*** *in Scheduling & Arbitration, each target j supports an atomic lock-acquire operation; the winner selected by Protocol C acquires the lock before commit, and at most one commit to j can linearize at τ_C in that round.*
4) ***Phase ordering via barriers:*** *memory barriers enforce the global order Read $\prec$ Compute $\prec$ Write $\prec$ Sanitization $\prec$ Scheduling/Arbitration $\prec$ Commit.*

6.2.0.1 Implementable primitives.: The model is implementable with hardware atomics and synchronization primitives: double-buffered register banks (or atomic transactions) for the Write linearization at τ_W, CAS/TAS-based per-target mutexes for lock acquisition, and release/acquire fences (or equivalent barrier instructions) between phases.

6.3 Interior Rewrite Rule (Indices $i \geq 2$)

Suppressing channel superscripts for brevity ($d_i = d_i^{(X)}$, $W_i = W_i^{(X)}$): if $d_i = +2$ then apply the exact local update:

$$d_i \leftarrow d_i - 2, \quad d_{i+1} \leftarrow d_{i+1} + 1, \quad d_{i-2} \leftarrow d_{i-2} + 1. \tag{4}$$

If $d_i = -2$ apply the symmetric update:

$$d_i \leftarrow d_i + 2, \quad d_{i+1} \leftarrow d_{i+1} - 1, \quad d_{i-2} \leftarrow d_{i-2} - 1. \tag{5}$$

These updates in (4) and (5) preserve the signed weighted sum $\sum_i d_i W_i$.

6.4 Boundary Rules and Lemmas

For $i = 0$: if $d_0 = +2$ then $d_0 \leftarrow d_0 - 2$, $d_1 \leftarrow d_1 + 1$ (symmetric for -2). For $i = 1$: if $d_1 = +2$ then $d_1 \leftarrow d_1 - 2$, $d_2 \leftarrow d_2 + 1$, $d_0 \leftarrow d_0 + 1$.

Lemma 6.2 (Boundary invariants)**.** *The boundary updates for $i = 0, 1$ preserve the weighted sum $\sum_i d_i W_i$ and cannot create an overflow at an index strictly greater than the current rightmost overflow index $R(d)$.*

Proof. Weighted-sum preservation follows by explicit substitution in the two boundary cases.

For $i = 0$ with overflow $d_0 = +2$, the update $d_0 \leftarrow d_0 - 2,\ d_1 \leftarrow d_1 + 1$ changes the weighted sum by

$$\Delta_0 = (-2)W_0 + (+1)W_1 = -2W_0 + W_1 = 0,$$

using the boundary-adapted identity $2W_0 = W_1$ (equivalently $W_{-2} = 0$ under virtual guards). The $d_0 = -2$ case is symmetric and gives $\Delta_0 = +2W_0 - W_1 = 0$.

For $i = 1$ with overflow $d_1 = +2$, the update $d_1 \leftarrow d_1 - 2,\ d_2 \leftarrow d_2 + 1,\ d_0 \leftarrow d_0 + 1$ changes the weighted sum by

$$\begin{aligned}\Delta_1 &= (-2)W_1 + (+1)W_2 + (+1)W_{-1} \\ &= -2W_1 + W_2 + W_{-1} = 0,\end{aligned}$$

where we use $2W_1 = W_2 + W_{-1}$ and the virtual-guard convention $W_{-1} = 0$. The $d_1 = -2$ case is symmetric and gives $\Delta_1 = +2W_1 - W_2 - W_{-1} = 0$.

Hence boundary rewrites preserve $\sum_i d_i W_i$.

For the rightmost-overflow claim, boundary updates only touch indices $0, 1, 2$ (and guards). If pre-round $R \geq 2$, no boundary action can affect any index > 2, so no new overflow can appear strictly to the right of R. If $R \in \{0, 1\}$, the boundary rule reduces the overflow at R and injects at most ± 1 within $\{0, 1, 2\}$, so no overflow can be created at any index $> R$. Therefore no boundary update creates an overflow strictly to the right of the pre-round $R(d)$. □

7 Sanitization and Deterministic Local Reconstruction

The physical pattern 11 is transient and must be sanitized. Sanitization is modeled by an operator S that maps $11 \mapsto 00$ and sets a reconstruction flag $K_i^{(X)}$. A deterministic reconstruction function $R_i^{(X)}$ examines a local neighborhood $d_{i-r}, \ldots, d_{i+r}$ ($r \leq 2$) and proposes a candidate injection $\tilde{v}_i \in \{-1, 0, 1\}$.

7.0.0.1 Timing convention for reconstruction input.: In each round, $R_i^{(X)}$ reads the *post-sanitization snapshot* (state after Write and Sanitization, before Scheduling & Arbitration), not the pre-sanitization Read snapshot; this post-sanitization view is latched once and shared consistently by all reconstructors for that round.

7.1 Protocol C: Formal Assumptions and Guarantee

Protocol C is the deterministic local arbitration policy used to resolve competing candidate injections targeting the same index.

7.1.0.1 Formal statement.: Given the candidate set targeting each index in one synchronous round, Protocol C deterministically selects and commits at most one candidate per target under Assumptions 1–5.

7.1.0.2 Assumptions (Protocol C).:

1) **Locality and snapshot consistency:** each candidate $v_{i \to j}$ is proposed by a reconstruction module whose source i satisfies $|i - j| \leq r$ for a fixed small radius r (typically $r \leq 2$); all such candidates in the round are computed from the same latched post-sanitization snapshot.
2) **Deterministic energy score:** each candidate is assigned a locally computable score $E_{i \to j}$ (energy-like) that depends only on the local neighborhood and is deterministic.
3) **Tie-breaking:** ties on $E_{i \to j}$ are broken deterministically (e.g., smaller source index). The deterministic winner selection (including tie-break resolution) is finalized before any target lock acquisition to avoid race conditions during commitment.

Actors: reconstructors in $N(j)$, arbiter j, lock j, commit stage for j.
(1) SCORE: each reconstructor computes score and proposal value from snapshot s_t.
(2) SELECT: arbiter j deterministically orders proposals by ($E_{r\to j}$, tie-break key) and emits winner id w_j.
(3) COMMIT: only w_j may execute lock acquisition. If acquisition succeeds, the winner commits at τ_C, then releases lock j; all non-winners are invalidated for the round.

Fig. 1. Protocol C per-target synchronization sequence (score, deterministic select, lock+commit).

4) **Atomic target lock:** committing a transfer to target j acquires an atomic local lock preventing a second commitment to the same j in the same round.
5) **Bounded candidate values:** each candidate $v_{i\to j} \in \{-1, 0, 1\}$.

7.2 Protocol C Synchronization Sequence and TOCTOU Avoidance

For each target j in one synchronous round, winner finalization is operationalized as a three-stage handshake that is local, deterministic, and race-safe.

7.2.0.1 Deterministic finalization mechanism.: The "winner finalized before lock acquisition" condition means that $\mathrm{winner}(j)$ is materialized as an immutable per-target record $(\mathrm{round_id}, j, w_j)$ after stage (2) and before any CAS on lock_j. This record may be produced by a local priority encoder, arbitration token ring, or deterministic arbiter FSM, provided all contenders for j are evaluated from the same snapshot and the same total order.

7.2.0.2 TOCTOU avoidance.: Time-of-check to time-of-use is avoided by binding check and use to the same round/target metadata: (i) proposals carry $(\mathrm{round_id}, j)$, (ii) lock acquisition validates that caller id equals w_j for the current $(\mathrm{round_id}, j)$, and (iii) commit re-validates unchanged $(\mathrm{round_id}, j, w_j)$ before linearizing at τ_C. Any stale or mismatched proposal is discarded and cannot commit.

Lemma 7.1 (Protocol C non-accumulation)**.** *Under Assumptions 1–5, Protocol C commits at most one transfer to any target j per round; hence the sum of committed transfers to j lies in $\{-1, 0, 1\}$.*

Proof. By Assumption 1, all contenders for a given target j originate within a bounded neighborhood. Each contender computes a deterministic score $E_{i\to j}$ (Assumption 2). The arbitration selects the contender with minimal $E_{i\to j}$ (Assumption 3), breaking ties deterministically. Once a winner is selected, the atomic target lock (Assumption 4) prevents any other contender from committing to the same j in the same round. Since each candidate value is in $\{-1, 0, 1\}$ (Assumption 5), the committed value is also in this set. Therefore at most one transfer is committed to j per round and the sum of committed transfers to j is in $\{-1, 0, 1\}$. □

7.2.0.3 Energy score template.: A concrete admissible score is:

$$E_{i\to j} := \alpha\, \Delta M^{\mathrm{local}}_{i\to j} + \beta\, |i-j| + \gamma\, \mathbf{1}_{\{K_i=0\}}, \qquad \alpha, \beta, \gamma > 0,$$

where

$$\Delta M^{\mathrm{local}}_{i\to j} := m_0 \sum_{k\in\mathcal{N}_r(j)} \Big(|d_k^{\mathrm{after}(i\to j)}| - |d_k^{\mathrm{before}}| \Big) W_k,$$

$\mathcal{N}_r(j) = \{k : |k-j| \le r\}$, and $\mathbf{1}_{\{K_i=0\}}$ penalizes candidates not sourced from a currently flagged sanitization site. A deterministic tie-break key (e.g., smaller source index i) completes the total order.

For fixed radius r, this score satisfies the five required properties: (1) *local computability* since all terms depend only on $\mathcal{N}_r(j)$, local flags, and local geometry; (2) *determinism* because the expression has no stochastic component; (3) *monotonicity in local mass change* because $\alpha > 0$ implies smaller $\Delta M^{\mathrm{local}}$ yields smaller score ceteris paribus; (4) *comparability/total ordering* by numeric score plus deterministic tie-break; and (5) *bounded evaluation cost* because evaluating the sums and penalties over fixed-size neighborhoods is $O(1)$ with respect to global length n.

7.2.0.4 Formal properties required of the local energy score $E_{i\to j}$.: The arbitration score $E_{i\to j}$ used by Protocol C must satisfy the following formal properties (the manuscript does not fix a single closed-form expression; implementations may choose a concrete formula that meets these requirements):

1) **Local computability:** $E_{i\to j}$ depends only on information available within the reconstruction radius r of the source i and the target j (local digits d_k for $|k-j| \le r$, local flags K_k, and deterministic tie-breakers). No global state or remote communication may be required to compute $E_{i\to j}$.
2) **Determinism:** given the same local context, $E_{i\to j}$ returns the same numeric value; randomness or non-deterministic heuristics are not permitted in the score computation.
3) **Monotonicity with respect to local mass change:** if candidate A yields a strictly smaller local increase in informational mass (as measured by ΔM restricted to the local neighborhood) than candidate B, then $E_A < E_B$ must hold, or the score must reflect a consistent preference that favors smaller local mass increases. This ensures arbitration favors energy-minimal reconstructions.
4) **Comparability and total ordering (with deterministic tie-breaker):** the score space must admit a total ordering; ties are resolved by a deterministic secondary key (for example, smaller source index i). This property guarantees a unique winner selection under Assumption 3 (Tie-breaking).
5) **Bounded evaluation cost:** computing $E_{i\to j}$ must require only $O(1)$ time with respect to the global word length n (i.e., cost depends only on the fixed radius r), so arbitration remains local and constant-time per target.

Any concrete implementation of Protocol C must document the chosen formula for $E_{i\to j}$ and demonstrate that it satisfies the five properties above.

8 Lexicographic Measure and Bounded Propagation

Define

$$R(d) := \max\{i : |d_i| = 2\}, \qquad m(d) := |\{i : |d_i| = 2\}|. \tag{6}$$

The lexicographic measure $p(d) = (R(d), m(d))$, defined in (6), is well-founded on finite configurations.

8.0.0.1 Complexity note.: Under the synchronous model and fixed reconstruction radius r, each target index performs constant-time local work per round (snapshot read, bounded-neighborhood score evaluation, deterministic tie-break, and at most one atomic commit), hence per-round computational cost is $O(1)$ per index and total per-round cost is $O(n)$ for word length n. The active working memory is linear in the number of indices (digit buffers, flags, and locks), i.e., $O(n)$ space. Let $m_0 := m(d^{(0)})$ be the initial overflow multiplicity. Since each round strictly decreases the lexicographic measure (R, m), the number of rounds is finite and bounded by a function $K(m_0)$; therefore total normalization cost is $O(n\,K(m_0))$. In the common regime where bounded-propagation yields a global constant round bound K, this simplifies to $O(n)$ time and $O(n)$ space per normalization.

Theorem 8.1 (Bounded propagation)**.** *Under synchronous read-then-write semantics, canonical inputs per channel, and Protocol C for reconstruction arbitration, any initial overflow event $|d_i| = 2$ is resolved within a neighborhood of radius $R_{\max}$ independent of the global length n. The number of synchronous rounds required to eliminate all ± 2 values is bounded by a constant K independent of n.*

Proof. We prove lexicographic descent of $p(d)$ per synchronous round. Consider one synchronous round executed according to the round semantics (Read, Compute, Write, Sanitization, Scheduling & Arbitration, Commit). Let $R := R(d)$ be the pre-round rightmost overflow index.

Interior rewrite effect. During the Compute phase, interior rewrites are computed for indices with $|d_i| = 2$. By the Fibonacci identity these rewrites remove the overflow at the acted index and inject at most ± 1 within the local neighborhood (indices $i-2, i, i+1$). By Definition 6.1, item 2 (Write linearization point), all Compute updates become visible only at the single linearization point τ_W; therefore no partial visibility or intra-round cascading can create a new overflow strictly to the right of the pre-round R.

Sanitization and arbitration effect. In the Sanitization phase, any transient 11 patterns are reset and flags $K_i^{(X)}$ are set. By the reconstruction timing convention, all candidates for Scheduling & Arbitration are computed from the same latched post-sanitization snapshot. In the Scheduling & Arbitration phase, Protocol C (Lemma 7.1) ensures at most one committed injection per target in the Commit phase. Therefore the combined effect of sanitization and committed compensations cannot produce a new overflow at an index $> R$ because each committed injection has magnitude at most ± 1 and is confined to the local radius r around its source; atomic commitment prevents multi-source accumulation at the same target in the same round.

Case analysis. If the interior rewrite at index R is applied in the Compute phase, the overflow at R is removed and any newly created overflows (if any) lie at indices $\leq R$. If the interior rewrite at R is not applied (e.g., because $R < 2$ and boundary rules apply), the boundary lemmas guarantee that either R decreases or the multiplicity $m(d)$ strictly decreases. In all cases $p(d)$ decreases lexicographically: either $R(d^+) < R(d)$ or $R(d^+) = R(d)$ and $m(d^+) < m(d)$.

Because $p(d)$ is well-founded on $\mathbb{N} \times \mathbb{N}$, repeated application of synchronous rounds terminates after finitely many steps within a bounded neighborhood. The existence of finite constants $R_{\max}$ and K follows; explicit numerical values are empirical consequences of exhaustive local enumeration (width-5) and the certified transition table. □

Lemma 8.2 (Cross-Channel Independence)**.** *With offsets $o_B \geq o_A + 1$, operations on channel A do not physically write to channel B. Normalization on A cannot force the creation of an overflow on B outside the bounded local neighborhood, provided the substrate enforces separate physical carriers per channel.*

9 Informational Mass–Energy Bookkeeping

To provide hardware designers with a register-level cost model, define an informational mass and a radiative energy register.

Definition 9.1 (Informational mass)**.** *Let $m_0 > 0$ be a model unit mass per unit positional weight. For a digit vector $V = (v_i)$ define the discrete informational mass in* (7)

$$M(V) := m_0 \sum_i |v_i|\, W_i. \tag{7}$$

Definition 9.2 (Radiative energy register)**.** *Let E_{rad} be a scalar register that accumulates informational energy produced by sanitization/annihilation events and supplies energy for reconstructions.*

9.0.0.1 Mass–energy operator.: Define the informational mass–energy model in (8)

$$\mathcal{E}(M) := Mc^2, \tag{8}$$

where c is a model parameter interpreted as the system refresh rate. The manuscript consistently treats c as a model parameter; any identification with the physical speed of light is explicitly left as an empirical hypothesis requiring calibration.

Lemma 9.3 (Specular energy symmetry)**.** *For any digit vector V with components in $\{-1, 0, 1\}$,*

$$M(-V) = M(V), \qquad \mathcal{E}(M(-V)) = \mathcal{E}(M(V)).$$

9.0.0.2 Operational energy consistency.: For each synchronous round the bookkeeping must satisfy (9):

$$\Delta\big(M_{\mathrm{tot}} c^2\big) + \Delta E_{\mathrm{rad}} = 0, \tag{9}$$

i.e., any decrease in discrete informational mass is recorded as an increase in the radiative energy register and vice versa.

9.0.0.3 Clarification (Informational mass versus physical mass).: The quantity $M(V)$ is an *informational bookkeeping mass*. The constants m_0 and c appearing in $\mathcal{E}(M) = Mc^2$ are **model parameters** chosen for dimensional bookkeeping. They are not a claim that the informational mass equals the invariant rest mass measured in particle physics experiments. Experimental mapping of m_0, c, and E_{rad} to substrate units is calibration-dependent and must be established per platform using controlled local events and fit-based parameter extraction. This bookkeeping is consistent with

thermodynamic analyses of irreversible computation [11], [12].

Remark 9.4 (Relation to relativistic negative-energy solutions)**.** *The SBA specular swap σ implements an informational inversion that is* formally *analogous to charge conjugation and parity inversion: it maps a signed informational configuration to its mirror image while preserving the informational mass M. SBA reproduces the* symmetry *between matter and antimatter at the level of information encoding; this is a representational correspondence, not a physical derivation.*

10 Discrete-to-Geometric Mapping: φ and π

The discrete Fibonacci index space admits natural geometric embeddings. Using Binet's formula and exponential radial sampling, the Fibonacci tiling can be associated with a logarithmic spiral whose growth factor is $\ln \varphi$. Geometric motifs (e.g., pentagonal arrangements) connect φ to π via $\varphi = 2\cos(\pi/5)$. These geometric observations motivate the choice of offsets o_A, o_B to avoid resonant overlap in implementations that map indices to spatial coordinates.

11 Worked Examples

Two concise worked examples illustrate normalization and sanitization dynamics.

11.1 Example 1: Simple Normalization

Channel X:

$$A^{(X)} = (0,0,1,0,0), \qquad A'^{(X)} = (0,0,1,0,0).$$

Intermediate:

$$d^{(X)} = (0,0,2,0,0).$$

Interior rewrite at index 2:

$$d_2 \leftarrow 0, \quad d_3 \leftarrow 1, \quad d_0 \leftarrow 1.$$

Result: $d^{(X)} = (1,0,0,1,0)$. Termination in one round.

TABLE 1
Round-by-round tracking for Example 1 (simple normalization).

Round/Phase	Local state evolution (channel X)
R1–Read	Snapshot: $d = (0,0,2,0,0)$.
R1–Compute	Overflow at index 2: schedule interior rewrite $d_2 \leftarrow d_2 - 2,\ d_3 \leftarrow d_3 + 1,\ d_0 \leftarrow d_0 + 1$.
R1–Write	Commit simultaneously: $d = (1,0,0,1,0)$.
R1–Sanitize	No 11 carrier event; flags remain zero.
R1–Sched/Arb	No flagged reconstructions; no candidates.
R1–Commit	No compensating injection; state unchanged.
R2–Read	$d = (1,0,0,1,0)$, with $\lvert d_i \rvert \le 1$ for all i: stable, terminate.

11.2 Example 2: Sanitization and Conflict Resolution

Channel X:

$$A^{(X)} = (0,1,0,0,0), \qquad A'^{(X)} = (0,1,0,0,0).$$

Intermediate:

$$d^{(X)} = (0,2,0,0,0).$$

Boundary rewrite at index 1 yields $d^{(X)} = (1,0,1,0,0)$. Suppose sanitization flags $K_0 = 1$ and $K_2 = 1$; candidate reconstructions produce $\tilde{v}_0 = +1, \tilde{v}_2 = 0$. Protocol C selects $v_0 = +1$, $v_2 = 0$. After applying compensating injection at next round, $d^{(X)} = (1,0,1,0,0)$ remains and normalization terminates.

TABLE 2
Round-by-round tracking for Example 2 (sanitization and arbitration).

Round/Phase	Local state evolution (channel X)
R1–Read	Snapshot: $d = (0,2,0,0,0)$.
R1–Compute	Boundary overflow at index 1: schedule boundary rewrite $d_1 \leftarrow d_1 - 2,\ d_2 \leftarrow d_2 + 1,\ d_0 \leftarrow d_0 + 1$.
R1–Write	Commit simultaneously: $d = (1,0,1,0,0)$.
R1–Sanitize	Assume transient carrier flags $K_0 = 1,\ K_2 = 1$.
R1–Sched/Arb	Candidates: $\tilde{v}_0 = +1,\ \tilde{v}_2 = 0$. Protocol C deterministically selects $v_0 = +1$, $v_2 = 0$.
R1–Commit	At most one commit per target; selected compensation is queued/committed for next-round semantics.
R2–Read/Write	After compensation handling, normalized state remains $d = (1,0,1,0,0)$, with $\lvert d_i \rvert \le 1$: stable, terminate.

12 Exhaustive Canonical Patterns (Complete List)

For reproducibility, Table 3 lists the complete set of canonical length-five patterns used in the exhaustive local enumeration. Each pattern is a 5-tuple in $\{-1,0,+1\}$ with non-adjacency. The enumeration yields exactly 43 patterns.

12.0.0.1 Verification of the canonical pattern table.: The canonical width-5 pattern table was produced by enumerating all 5-tuples in $\{-1,0,1\}^5$ and filtering by the non-adjacency constraint $x_i x_{i+1} = 0$. As a verification step we (i) checked that the table contains exactly 43 distinct entries, (ii) confirmed that each listed 5-tuple satisfies the non-adjacency constraint, and (iii) validated that the union of the listed patterns equals the filtered enumeration set (no omissions or duplicates). These checks were performed deterministically and exhaustively on the finite width-5 search space; the aggregated counts reported in Section 12 follow directly from this verified table.

13 Exhaustive Pair Enumeration and Critical Pairs

Given the 43 canonical patterns per channel, ordered pairs of canonical patterns yield at most $43 \times 43 = 1849$ ordered pairs. The conceptual exhaustive check classifies ordered pairs according to whether they produce a center overflow $\lvert d_{\text{center}} \rvert = 2$ at the initial intermediate array.

For the offsets $o_A = 2, o_B = 3$ and under canonical inputs, the enumeration yields exactly **162 critical ordered pairs** for which the center sum equals ± 2 and sanitization is invoked. Table 4 reports the aggregated distribution of critical pairs by patternA id (left operand).

TABLE 3
Canonical Patterns of Length Five (43 Patterns)

ID	Pattern
1	$(-1, 0, -1, 0, -1)$
2	$(-1, 0, -1, 0, 0)$
3	$(-1, 0, -1, 0, 1)$
4	$(-1, 0, 0, -1, 0)$
5	$(-1, 0, 0, 0, -1)$
6	$(-1, 0, 0, 0, 0)$
7	$(-1, 0, 0, 0, 1)$
8	$(-1, 0, 0, 1, 0)$
9	$(-1, 0, 1, 0, -1)$
10	$(-1, 0, 1, 0, 0)$
11	$(-1, 0, 1, 0, 1)$
12	$(0, -1, 0, -1, 0)$
13	$(0, -1, 0, 0, -1)$
14	$(0, -1, 0, 0, 0)$
15	$(0, -1, 0, 0, 1)$
16	$(0, -1, 0, 1, 0)$
17	$(0, 0, -1, 0, -1)$
18	$(0, 0, -1, 0, 0)$
19	$(0, 0, -1, 0, 1)$
20	$(0, 0, 0, -1, 0)$
21	$(0, 0, 0, 0, -1)$
22	$(0, 0, 0, 0, 0)$
23	$(0, 0, 0, 0, 1)$
24	$(0, 0, 0, 1, 0)$
25	$(0, 0, 1, 0, -1)$
26	$(0, 0, 1, 0, 0)$
27	$(0, 0, 1, 0, 1)$
28	$(0, 1, 0, -1, 0)$
29	$(0, 1, 0, 0, -1)$
30	$(0, 1, 0, 0, 0)$
31	$(0, 1, 0, 0, 1)$
32	$(0, 1, 0, 1, 0)$
33	$(1, 0, -1, 0, -1)$
34	$(1, 0, -1, 0, 0)$
35	$(1, 0, -1, 0, 1)$
36	$(1, 0, 0, -1, 0)$
37	$(1, 0, 0, 0, -1)$
38	$(1, 0, 0, 0, 0)$
39	$(1, 0, 0, 0, 1)$
40	$(1, 0, 0, 1, 0)$
41	$(1, 0, 1, 0, -1)$
42	$(1, 0, 1, 0, 0)$
43	$(1, 0, 1, 0, 1)$

TABLE 4
Distribution of critical pairs by patternA id (counts).

ID	Count	ID	Count
1, 2, 3	9 each	25, 26, 27	9 each
9, 10, 11	9 each	33, 34, 35	9 each
17, 18, 19	9 each	41, 42, 43	9 each
Total: 162 critical pairs			

13.0.0.1 Reproducibility statement: The enumeration reported above is the result of an exhaustive local analysis of width-5 canonical windows under the stated offsets. The explicit ordered pair list and the certified transition table are available as supplementary material upon request to the author; the manuscript documents the method and the aggregated counts used in the formal arguments.

13.1 Numerical Extraction of $R_{\max}$ and K from Width-5 Enumeration

Let $\mathcal{T}$ denote the certified finite transition table produced by exhaustive width-5 enumeration under fixed offsets and Protocol C assumptions. Each row $\tau \in \mathcal{T}$ corresponds to one local seed configuration and records its synchronous-round evolution to a stable state (no $|d_i| = 2$).

13.1.0.1 Operational definitions on $\mathcal{T}$.: For each transition row τ:

- let $i_{\text{seed}}(\tau)$ be the seed index where the initial overflow/reconstruction event is centered;
- let $S_t(\tau)$ be the active index set after round t (indices touched by overflow/reconstruction effects);
- let $T(\tau)$ be the first round such that the local state is stable.

Define

$$R_{\max} := \max_{\tau \in \mathcal{T}} \max_{0 \le t \le T(\tau)} \max_{i \in S_t(\tau)} |i - i_{\text{seed}}(\tau)|,$$

$$K := \max_{\tau \in \mathcal{T}} T(\tau).$$

Thus $R_{\max}$ is the worst observed spatial spread and K is the worst observed round latency over the finite exhaustive set.

13.1.0.2 Deterministic computation procedure.: (1) Enumerate all canonical width-5 operand pairs (per channel) and construct $\mathcal{T}$. (2) Simulate synchronous rounds exactly as specified (Read/Compute/Write/Sanitize/Schedule/Commit) until stability for each row. (3) Record $T(\tau)$ and the maximum

displacement from $i_{\text{seed}}(\tau)$. (4) Take global maxima to obtain $R_{\max}$ and K.

13.1.0.3 Reproducibility note.: Because $\mathcal{T}$ is finite, the procedure yields exact numerical values (not asymptotic estimates) once the certified transition table is materialized.

14 Notes on Experimental Prototyping

This section provides conceptual design considerations for implementers.

14.1 Substrate Primitives

- **Carrier implementation:** carriers may be realized as pairs of photonic modes, electronic states, or DNA base pairs. Implementations must ensure deterministic sanitization of the transient 11 state.
- **Local read/write rounds:** substrate must support synchronous read-then-write cycles or an equivalent atomic update primitive.
- **Atomicity and local locking (operational requirement):** the substrate must provide an atomic write primitive for committing the results of the Compute phase to the physical carriers and must support a local, per-target locking mechanism for arbitration. Concretely, the hardware or microcontroller layer must guarantee that (i) all writes in the Compute phase become visible simultaneously at the end of the Write phase (no partial visibility or intra-round cascading), and (ii) a target-level atomic lock can be acquired and released within a single round to prevent multiple concurrent commitments to the same target index. These primitives are required to implement the synchronous read-then-write semantics and the Protocol C arbitration assumptions used in the proofs.
  Implementation note: a per-target latch bank plus CAS-style lock bit (or a tiny local arbiter FSM) is sufficient. Resolve winner selection from a latched candidate set first, then perform a single lock-acquire-and-commit transaction so no select/lock race can occur.

- **Arbitration and energy registers:** Protocol C requires deterministic arbitration and a locally accessible energy register E_{rad} per neighborhood.
- **Operational enforcement of cross-channel independence:** hardware must enforce separate physical carriers for channels A and B or equivalent access control to prevent cross-writes.

14.2 Measurement and Validation

- **Functional verification:** exhaustive width-5 window tests to verify bounded propagation.
- **Robustness tests:** inject transient 11 patterns and verify deterministic reconstruction and non-accumulation under worst-case overlapping flags.
- **Calibration protocol (conceptual):** to map m_0 and c to substrate energy units, measure per-sanitization energy and per-reconstruction energy under controlled canonical inputs and fit model parameters accordingly.

15 Discussion, Limitations, and Future Work

- **Limitations:** the framework assumes synchronous rounds and deterministic local arbitration primitives. Asynchronous implementations require additional formal analysis. The mass–energy bookkeeping is a modeling device; mapping m_0 and c to physical energy units depends on the substrate and must be calibrated experimentally.
- **Future work:** formal machine-checked verification (Coq/Isabelle) of the bounded-propagation invariant, hardware prototypes implementing Protocol C, integration experiments with archival DNA layers, and thermodynamic calibration experiments inspired by irreversible-computation bounds are natural next steps [11], [29], [30].

15.1 Scalability Challenges

Scaling SBA from local windows to very large arrays introduces three coupled bottlenecks:

- **Synchronization pressure:** global round barriers become progressively expensive with increasing array diameter. Even when per-index work is $O(1)$, barrier latency and clock skew can dominate wall-clock throughput for large n.
- **Arbitration fan-in hotspots:** nonuniform input statistics can create spatially clustered flagged regions, increasing contention on target locks and local arbiters. This does not violate bounded propagation, but it can degrade effective throughput through repeated local conflicts.
- **Memory-bandwidth locality limits:** the model is local in logic, but practical implementations still require moving digit buffers, flags, and metadata. If physical placement does not preserve neighborhood affinity, interconnect traffic can dominate compute.

Practical scaling therefore requires hierarchical tiling: neighborhood-local schedulers, region-level asynchronous handshakes, and periodic global synchronization only for consistency checkpoints. This architecture preserves local correctness while reducing barrier frequency and contention amplification.

15.2 Thermal Noise Analysis (Model-Level)

For noisy substrates, let observed local digit state be $\hat{d}_i = d_i + \eta_i$, with η_i representing thermal/noise-induced perturbations in carrier readout or thresholding. A useful abstraction is a temperature-dependent local error probability p_T that a valid ternary carrier state is misclassified in one round. Under independent first-order approximation:

$$\mathbb{E}[N_{\text{false flag}}] \approx n\, p_T, \qquad \mathbb{E}[N_{\text{extra rounds}}] \propto p_T,$$

so false sanitization flags increase linearly with size for fixed p_T. Protocol C still enforces non-accumulation per target, but

elevated p_T increases arbitration load and energy-register activity. Design implications are: (i) adaptive thresholds tied to online noise estimates, (ii) temporal majority filtering for flag generation, and (iii) local confidence tagging so low-confidence candidates are penalized in the deterministic score. These mitigations are consistent with irreversible-computation and information-thermodynamics constraints [11], [29], [30].

15.3 Plan for Quantum Validation

Given the manuscript's explicit distinction between informational formalism and physical claims, a staged validation path is appropriate:

1) **Classical baseline certification:** verify finite-state transition correctness and measured $(R_{\max}, K)$ on deterministic hardware emulation.
2) **Quantum-circuit embedding of local rules:** encode one SBA neighborhood update as reversible subcircuits (unitary blocks plus measurement for sanitization events), then compare transition statistics with classical ground truth [28], [31].
3) **Noise-channel sensitivity study:** run the embedded circuit under depolarizing/dephasing noise models and estimate an effective p_T-analog for quantum hardware, relating logical error rates to added normalization latency.
4) **NISQ-device pilot:** implement minimal instances on available noisy intermediate-scale quantum platforms, emphasizing reproducibility and uncertainty quantification rather than speedup claims [32], [33].
5) **Cross-platform falsifiability criteria:** predefine acceptance tests (state-transition fidelity thresholds, bounded-propagation preservation, and energy/bookkeeping consistency proxies) to separate model-valid behavior from hardware artifacts.

This plan keeps the framework scientifically conservative: quantum experiments are used to test representational and dynamical consistency, not to assert unproven physical equivalence.

15.3.0.1 Experimental disclaimer: Predictions that connect the informational bookkeeping to measurable physical effects are speculative and require dedicated experimental protocols. The present framework supplies a theoretical mapping; it does not constitute empirical proof of cosmological claims.

16 Conclusion

We presented a substrate-agnostic, mathematically rigorous framework for a Specular Bit Architecture intended as an informational layer for bio-photonic neuromorphic chips and DNA storage systems. The framework provides canonical encoding semantics, exact local rewrite rules based on Fibonacci positional weights, deterministic sanitization and arbitration policies, a lexicographic termination measure, and an informational mass–energy bookkeeping model. The manuscript explicitly separates formal informational claims from speculative physical interpretations and provides a clear roadmap for reproducibility and formal verification.

Appendix A
Canonical Width-5 Patterns

For reproducibility, Table 3 lists the complete set of canonical length-five patterns used in the exhaustive local enumeration. Each pattern is a 5-tuple in $\{-1, 0, +1\}$ with non-adjacency.

Appendix B
Exhaustive Pair Enumeration and Critical Pairs (Notes)

Given the 43 canonical patterns per channel, ordered pairs of canonical patterns yield at most $43 \times 43 = 1849$ ordered pairs. The conceptual exhaustive check classifies ordered pairs according to whether they produce a center overflow $|d_{\text{center}}| = 2$ at the initial intermediate array. The aggregated counts are reported in Table 4.

Appendix C
Appendix: Lemma List for Formal Verification

The following lemmas are recommended for machine-checked formalization:

1) Zeckendorf uniqueness for shifted Fibonacci weights.
2) Preservation of weighted sum by interior and boundary rewrites.
3) Protocol C non-accumulation lemma under stated assumptions (Lemma 7.1).
4) Lexicographic descent lemma for one synchronous round.
5) Bounded propagation theorem (termination from lexicographic descent, Theorem 8.1).

Appendix D
Appendix: Numerical Bounds Remark

Explicit numerical bounds for $R_{\max}$ and K are empirical consequences of the exhaustive width-5 enumeration and the certified transition table. Concretely, $R_{\max}$ is obtained as the maximum observed spatial displacement of any overflow/reconstruction effect from its seed index across all enumerated canonical window transitions, while K is the maximum number of synchronous rounds to reach a $|d_i| \leq 1$ stable state over the same finite transition set. The abstract lexicographic argument used in Theorem 8.1 guarantees existence but not numerical values; the reported counts and transition table provide the empirical evidence for concrete bounds. Practically, these bounds define the minimum local neighborhood radius and worst-case local latency that hardware schedulers must provision, and they are valid under the stated width-5 locality model and Protocol C assumptions.

Appendix E
Appendix: Extended Proof Sketch for Bounded Propagation

This appendix provides a structured expansion of the argument behind Theorem 8.1, with hypotheses aligned to the formal execution model in Definition 6.1 and arbitration constraints in Lemma 7.1.

E.0.0.1 Hypotheses.: Assume (H1) finite-support initial intermediate state $d^{(0)}$, (H2) synchronous read-then-write rounds with a global write linearization point (Definition 6.1), (H3) local rewrite legality from the Fibonacci identity $2W_i = W_{i+1} + W_{i-2}$, (H4) sanitization followed by deterministic Protocol C arbitration with per-target non-accumulation (Lemma 7.1), and (H5) fixed reconstruction radius r.

E.0.0.2 Phase-wise invariant flow.: Within one round, the Compute phase can only schedule updates over bounded neighborhoods of currently overflowing indices. Because visibility is deferred to the Write linearization point, no intra-round chain reaction can travel rightward via partially committed writes. Sanitization maps transient ambiguous microstates to canonical carrier states and may emit bounded-magnitude reconstruction candidates, but Protocol C admits at most one committed transfer per target and therefore blocks same-round multi-source accumulation.

E.0.0.3 Lexicographic descent restated.: Let $p(d) = (R(d), m(d))$ with $R(d) = \max\{i : |d_i| = 2\}$, $m(d) = |\{i : |d_i| = 2\}|$. For any non-stable round transition $d \mapsto d^+$, either the rightmost overflow is eliminated with no creation beyond $R(d)$, so $R(d^+) < R(d)$, or $R(d^+) = R(d)$ and the multiplicity strictly decreases, $m(d^+) < m(d)$. Thus $p(d^+) <_{\text{lex}} p(d)$.

E.0.0.4 Termination and bounded locality.: Since p takes values in $\mathbb{N} \times \mathbb{N}$ with the well-founded lexicographic order, infinite descending chains are impossible; normalization terminates after finitely many rounds. Because each round touches only neighborhoods determined by rewrite stencils and fixed radius r, total spatial spread from any seed event is bounded by a finite radius independent of global length n. This yields existence of finite constants $R_{\max}$ and K, while their numerical values are extracted empirically from the certified finite transition table.

Appendix F
Appendix: Dissipative Structures and Informational Entropy

This appendix refines the interpretation of the SBA dynamics as an open informational process with local dissipation, while preserving the model-level disclaimer of Section "Informational Mass–Energy Bookkeeping."

F.0.0.1 Entropy functional on local states.: For a local neighborhood $\mathcal{N}$, let q_s denote the empirical frequency (or round-ensemble weight) of microstate s. Define the informational entropy

$$\mathcal{H}(\mathcal{N}) := -\sum_s q_s \log q_s.$$

Transient ambiguous patterns (including unsanitized 11 carriers) increase local configurational uncertainty; deterministic sanitization and arbitration reduce this uncertainty by contracting admissible post-round state sets.

F.0.0.2 Dissipative interpretation.: Under the bookkeeping constraint

$$\Delta(M_{\text{tot}}c^2) + \Delta E_{\text{rad}} = 0,$$

local decreases in informational mass correspond to positive updates of E_{rad}. In this sense, normalization behaves as a dissipative mechanism: logically disallowed or unstable local configurations are removed, and their modeled energetic counterpart is exported to a local register. The process is structurally analogous to dissipative-structure narratives in non-equilibrium thermodynamics (ordered macroscopic behavior sustained by local dissipation) [10], but remains an informational analogy rather than a direct physical identity claim.

F.0.0.3 Steady operational regime.: When input statistics and round scheduling are stationary, the architecture approaches a statistical steady regime where average injected ambiguity is balanced by average sanitized/reconstructed dissipation. In that regime, bounded propagation (Theorem 8.1) ensures that entropy-producing corrections remain local, preventing system-wide cascades and supporting scalable composition across independent channels.

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Marco Oppido Marco Oppido is an independent researcher based in Rome, Italy. His research interests span theoretical physics, informational thermodynamics, and unconventional computing architectures. He is actively involved in developing theoretical frameworks that integrate quantum mechanics, probability fields, and dissipative structures into novel computational paradigms. His current work focuses on the Specular Bit Architecture (SBA), exploring its applications in neuromorphic photonics, bio-electronic microcircuits, and DNA data storage. Bridging the gap between fundamental physics and computer arithmetic, his research aims to establish rigorous mathematical primitives for next-generation, substrate-agnostic information processing.